PROMISE THEORY AND A SOCIETY OF AGENTS

The interaction between ACME's reimbursement agent and the partner's payment agent illustrates a broader pattern that emerges when independent systems cooperate. When independent systems work together, control is rarely centralized. Each organization manages its own systems through local policies and operational controls. Promise theory (https://markburgess.org/promises.html) offers a helpful way to understand how cooperation can still develop in such environments. Instead of one system commanding another, each participant makes promises about how it will behave and others decide whether those promises are enough to rely on.

In the reimbursement example in this section, ACME and its financial partner operate under legal agreements that define their business relationship. These agreements outline responsibilities, liabilities, and audit requirements. In addition, each organization agrees to specific protocols, schema, and other technical standards. Within this framework, the software agents representing each organization make more specific operational commitments. For instance, the partner's payment agent may promise to process only reimbursements below a certain threshold, require receipts for each payment, and return signed records documenting every transaction.

These promises are significant because they are based on policies that limit the agent's behavior within its own domain. The partner's Cedar policies guarantee that the payment agent cannot exceed the limits specified in its promise, while ACME's policies specify when it is appropriate to delegate authority.

Over time, each organization observes whether the other's agents keep their promises. The results of these interactions build a reputation database that affects future delegation choices. An agent that consistently fulfills its commitments becomes a more trusted partner, while one that violates expectations quickly loses the ability to receive delegated authority.